Recommended Assessment

Stepper Motor Lab

Stepper Motor Drive Modes

1. Complete the following tables for how to drive the motor in each of these drive modes assuming a clockwise rotation starting with A. Use 0's and 1's for the coil being off and on respectively.

Wave Drive			
A+	A-	B+	B-

Full Step Drive			
A+	A-	B+	B-

Half Step Drive			
A+	A-	B+	B-

2. According to the motor's datasheet (located in the stepper folder), what is the step angle of the motor? Based on this, how many steps are needed to complete a full rotation if we are using wave drive? If we are using full step drive? If we are using half step drive?

3. How would you make your motor take 2 turns while maintaining the 1 rotation per second with full step drive? What if you wanted the 2 turns in just 1 second?

4. What happened when you reduced the stepper amplitude in all 3 cases to 0.2? What did you think was going to happen?

5. What happened when changing the frequency? What did you notice?

6. What logic did you use when using **`fullStepDrive`** at a frequency of 400 to have one rotation per second? How did you do the conversion? What if you wanted 17 steps/second or 270?

Microstepping

7. What is the relationship between the waveforms applied to **`a_plus`** and **`a_minus`**?

8. With a `frequency` of 500 and an **`angularFrequency`** of 30, as set in the file, how many microsteps are in each half wave? How would you calculate this?

9. Based on the **`angularFrequency`** of the sine wave, how many seconds should it take to do 1 full rotation? How would you calculate this?

10. What happens when the angular frequency is increased? What about when the frequency of the file is decreased? Why?

11. What pattern did you see when turning the motor clockwise vs counterclockwise? What did you have to change?

12. If we wanted at least 8 samples per period of the sine wave, what is the largest angular frequency we can go to (to the nearest 10) if we keep the 500Hz of the python file?

13. How does the body of the motor feel when using this angular frequency when compared to a smaller angular frequency? Try stalling the motor by holding onto the shaft, can you easily do it? What about when decreasing the amplitude to 0.2?